Research on the distribution and control mechanism of evapotranspiration in the Nanxiaohegou watershed based on an improved S–W model

YANG Nan[1] , SONG Xiaoyu[2] , DENG Jianwei[1] , LI Lanjun[2] , ZHAO Xinkai[2] , MENG Pengfei[2] , FU Chong[2],WEI Wanyin[2], ZHANG Yubin[1] , DING Lin[1] , LI Haolin[3]

1. Gansu Academy for Water Conservancy, Lanzhou 730000, Gansu, China;

2. State Key Laboratory of Eco-Hydraulics in Northwest Arid Region of China, Xi'an University of Technology, Xi 'an 710048, Shaanxi, China;

3. Beijing University of Technology, Beijing 100124, China.

*correspondencing author SONG Xiaoyu. Email: songxy@xaut.edu.cn

Abstract

Accurate quantification of evapotranspiration (*ET*) and its components, and identifying their controlling factors, were beneficial for the rational evaluation, planning, and management of regional water resources. Based on continuous long-term observation data and field experiments conducted from 2016 to 2020 in the Nanxiaohegou watershed, a typical small watershed for water and soil conservation on the Loess Plateau, an improved Shuttleworth-Wallace (S-W) model was used to simulated the dynamic changes of *ET* and its components in typical plantation land. Additionally, we analyzed the coupling relationships between plant transpiration (*T*), soil evaporation (*E*), and control factors using Structural Equation Modeling (SEM). The results showed that: (1) The improved S–W model had good applicability for evaluating *ET* and its components in Nanxiaohe Watershed. The threshold value of soil surface resistance (r_s^s) was 50–2500 $s \cdot m^{-1}$, exhibiting an exponential relationship with the empirical function of soil surface water content (θ); moreover, the larger the sand content in soil particles, the greater the slope of the line. (2) *ET* ranged from 276.76 mm to 402.86 mm in typical plantation land, with multi-year average contributions of *T* and *E* accounting for 51.6% and 48.4% of *ET*, respectively. The patterns of *ET*, *T*, and *E* showed no significant monthly changes but drastic fluctuations during the day. The fluctuation trends of T and E largely reflected annual precipitation patterns but lagged behind rainfall. (3) SEM analysis revealed that net radiation (R_n), air temperature (T_a), and θ had the most significant impacts on *ET*, with R_n having the largest impact on *T* (total impact of 0.614) and T_a having the greatest impact on *E* (total impact of 0.426). *T* was positively correlated with *E*, with a contribution coefficient reaching 0.503. Evaluating *ET* and its components based on an improved S–W model established a foundation for deeply revealing ecological and hydrological processes in arid and semiarid regions.

Key words Evaporative components; Improve the S-W model; SEM Model; Evaporation resistance; Loess Plateau; Ecological hydrological processes

1. Introduction

Evapotranspiration (*ET*) is an indispensable part of the hydrological cycle[1], playing a significant role in the water and energy balance of ecosystems. Particularly in the Loess Plateau region, *ET* accounts for more than 80% of the water loss in the system[2]. As a key factor in plant water requirements and water resource evaluation, *ET* is also an important indicator for drought monitoring. Its two components, plant transpiration (*T*) and soil evaporation (*E*), involve different biological and physical processes and have distinct dominant factors[3]. Therefore, accurately simulating *ET* and its components and studying the control mechanisms of evapotranspiration distribution are significant for the efficient planning, management, and application of water resources in the loess gully region.

Currently, *ET* assessment methods mainly include stable isotopes, sap flow, and eddy covariance techniques[4], which have the advantage of direct measurement without being constrained by hypothetical conditions[5]. However, they are limited in practical application becuse of high economic costs, low observation frequencies, difficulties in instrument maintenance, and narrow applicability. Hence, mathematical modeling is the primary method for *ET* assessment, with the Penman-Monteith model (P-M model) [6-7] and the Shuttleworth-Wallace model (S-W model) [8-9] being widely used for their simplicity, adaptability, and cost-effectiveness, especially in areas with complex geomorphological conditions. However, the P-M model cannot separate the components of evapotranspiration, whereas the The S-W model is better due to its ability to capture physical processes, fewer parameters with clear physical meanings, and higher simulation accuracy[10]. The improved S-W model can address the issue of calculating soil surface resistance for different soil textures[11-12] and has been proven effective in *ET* separation applications in arid and semi-arid areas.

Regarding the control mechanisms of *ET* distribution and the interrelationships among its factors, scholars worldwide often analyze these through comparing pairwise correlation coefficients or using traditional statistical methods like multiple regression[12,14-15]. These methods only analyze the relationship between one or more independent variables and one dependent variable, failing to assess multiple independent variables' impact on multiple dependent variables. They also often overlook the autocorrelation between independent and dependent variables, leading to inconsistent or even opposite results[16-17]. In recent years, Structural Equation Modeling (SEM) has emerged as an effective method to address above issues. SEM not only fully considers the correlation between

independent variables but also distinguishes the impact of independent variables on dependent variables, which has been widely applied in economics and ecology[18], although its application in analyzing the control factors of evapotranspiration is still limited.

As a typical prototype observation watershed in the loess gully area, this study uses long-term continuous observation data and field experiments from the Nanxiaohegou Watershed to quantify and dynamically evaluate *ET* and its components through the improved S-W model. The SEM model is used to analyze the control mechanisms, clarifying the response process between *ET* and its components and control factors. This provides a reference for estimating plant water requirements, improving water use potential, and conducting drought monitoring and assessment research in the loess gully area of the Loess Plateau.

2. Materials and Methods

2.1 Overview of the study area

The study area was located in the Nanxiaohegou Watershed, within the Xifeng District of Qingyang City, Gansu Province (35°40′43"~35°44′58"N, 107°30′8"~107°37′52"E), covering an area of 38.7 km^2. This watershed, established in 1951 by the Xifeng Station of the Loess Plateau Soil and Water Conservation Experiment, served as a typical prototype observation site for the gully region of the Loess Plateau (Figure 1). It had the distinctive topographical features of the Loess Plateau gully region, including tablelands, ridges, and valleys. The geological structure of the watershed was simple, with sandy loam soil that was prone to erosion and water loss. In recent years, with the implementation of large-scale soil and water conservation efforts in the Loess Plateau region, the vegetation coverage in the watershed had significantly increased. The main vegetation types included black locust (*Robinia pseudoacacia*), Oriental arborvitae (*Platycladus orientalis*), Chinese pine (*Pinus tabulaeformis*), Siberian apricot (*Armeniaca sibirica*), and alfalfa (*Medicago sativa*).

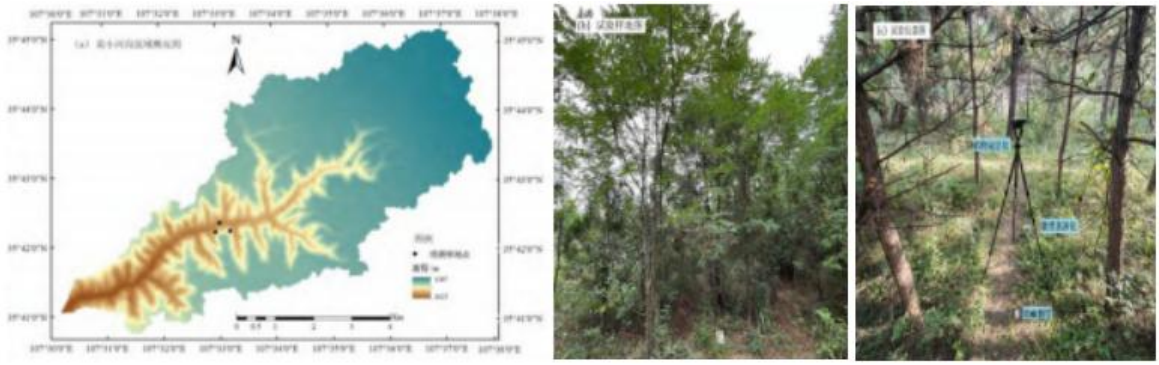


Fig 1. Schematic diagram of study area and plot and partial test map

2.2 Data sources and processing

Three sample plots with consistent tree ages and good growth were selected for the mixed plantations of black locust, Oriental arborvitae, and Chinese pine. Field experiments and observations were conducted within these plots from May to October in the years 2016, 2017, 2018, 2019, and 2020. The observations primarily included meteorological elements, soil moisture content, soil evaporation, and vegetation leaf area index. The plant parameters were represented by the average values of the three tree species, and the specific conditions of the tree species are presented in Table 1.

Table 1. Basic information of the test site

树种	树高/cm	胸径/cm	冠幅/cm	土质	地形	坡位	坡度/（°）
刺槐	564.11±29.60	15.91±2.42	293.11±15.20	砂壤土	阶地	中	26
侧柏	499.43±27.91	17.44±3.73	211.42±11.63	砂壤土	阶地	中	22
油松	446.21±24.23	15.41±1.90	191.60±12.31	砂壤土	阶地	中	21

2.2.1 Meteorological data

Meteorological data were measured by three small field weather stations (Watch dog series 2000, China). The instruments, powered by solar energy, recorded indicators such as wind speed, wind direction, air temperature, humidity, total radiation, and precipitation. The weather stations were installed in open areas near the three sample plots, with an observation interval set at 1 hour. The observation results were converted into daily-scale data for subsequent analysis and calculation. To avoid missing or distorted data, the weather stations were checked and maintained approximately every week. Any missing data were rare and were replaced by data from the nearest weather station.

2.2.2 Soil moisture content

Soil moisture content was measured using TDR (Trime TDR, Germany) probes. Experiments were conducted on the three sample plots, with two measurement points at each plot, resulting in a total of six TDR probes. The probes were buried at a location equivalent to 0.7 times the average radius of the canopy. During the installation of the probes, efforts were made to minimize disturbance to the original soil of the plantations. The TDR probes measured soil moisture at depths of 0 to 2 meters, with measurement intervals of 10 cm from 0 to 1 meter and 20 cm from 1 to 2 meters. The observation interval ranged from 1 to 3 days, and the average values were calculated. To reduce errors, observations were conducted between 8:00 and 11:00 AM each day, with three measurements at each depth in each probe. The soil moisture content of each soil layer was determined by taking the average of the three observations. Additional measurements of soil moisture content were conducted after each rainfall event.

2.2.3 Soil evaporation

Soil evaporation was measured using micro-lysimeters. The observations were conducted on the three sample plots, with three measurement points at each plot, resulting in a total of nine PVC tubes with a length of 25 cm and a diameter of 16 cm. During installation, the PVC tubes were first driven vertically into the soil using a soil tamper, and then the original soil was carefully removed, sealed with gauze at the bottom, and placed back in position, ensuring that the soil surface inside the micro-lysimeter was level with the tube opening. Vegetation on the soil inside the PVC tubes was removed. A wire handle was attached to the top for easy subsequent observations. Observations were conducted at 18:00 every day, with an interval of 1 to 3 days. Actual evapotranspiration was calculated using the weighing method with an electronic scale of 0.01 g precision. The difference between two weighings was considered as the soil evaporation rate during a certain period. To ensure the measurement accuracy of the micro-lysimeters, the soil should be changed every 20 days, and the mass of the PVC tube and the soil inside should be consistent with the mass of the previous observation when changing the soil.

2.2.4 Leaf area index

Leaf Area Index (LAI) was measured using a plant canopy observation system (Winscanopy 2006, Canada). The measurements were conducted on the three sample plots, with two fixed observation points at each plot. After leveling the tripod, a "fisheye" camera was used to photograph three to five representative trees. The photos were processed in the Winscanopy software to obtain the LAI. The observation interval was 4 to 7 days, and the average value was calculated.

2.3 Research Methods

2.3.1 S-W model

The Shuttleworth-Wallace model (S-W model) assumed a uniform plant canopy coverage and introduced canopy resistance and soil resistance parameters. It considered evapotranspiration as originating from two "sources": soil evaporation and plant transpiration. The total water vapor source could be represented by the latent heat flux λ*E*, and its calculation formula was as follows [10]:

$$\lambda ET = \lambda E + \lambda T = C_S PM_S + C_C PM_C \quad (1)$$

in the formula,

$$E = PM_s = \frac{\Delta R + \dfrac{\rho C_p D - \Delta r_a^c \left(R - R_s\right)}{r_a^a + r_a^s}}{\Delta + \gamma\left(1 + \dfrac{r_s^s}{r_a^a + r_a^s}\right)} \quad (2)$$

$$T=PM_C=\frac{\Delta R+\frac{\rho C_p D-\Delta r_a^c R_s}{r_a^a+r_a^c}}{\Delta+\gamma\left(1+\frac{r_s^c}{r_a^a+r_a^c}\right)} \quad (3)$$

In the formula: *ET* represented the simulated evapotranspiration rate (mm d^{-1}); λ was the latent heat of vaporization of water (MJ kg^{-1}); PM_s was the simulated soil evaporation rate (mm·d^{-1}); PM_c was the simulated plant transpiration rate (mm·d^{-1}); C_s and C_c were the proportionality coefficients for soil evaporation and plant transpiration, respectively; r_s^s was the soil surface resistance (s·m^{-1}); r_a^s was the aerodynamic resistance from the soil surface to the canopy height for latent heat (s·m^{-1}); r_s^c was the stomatal resistance of the plant canopy (s·m^{-1}); r_a^c was the aerodynamic resistance from the leaf surface to the average height of the canopy (s·m^{-1}); r_a^a was the aerodynamic resistance from the canopy height to the reference height for latent heat (s·m^{-1}); Δ was the slope of the saturation vapor pressure curve with temperature (kPa·℃$^{-1}$); *P* was the air density (1.29 kg·m^{-3}); C_p was the specific heat of air at constant pressure [1.03×10^{-3} J ·(kg · ℃)$^{-1}$]; *D* was the air saturation vapor pressure difference (kPa); γ was the psychrometric constant (0.67 hPa·k^{-1}).

In the formula: *R* represented the radiation absorbed by the canopy (W·m^{-2}), and R_S represented the radiation absorbed by the soil (W·m^{-2}). The calculation formula was as follows:

$$R = R_n - G \quad (4)$$

$$R_S = R_{ns} - G \quad (5)$$

$$R_{ns} = R_n exp(-C\mathrm{LAI}) \quad (6)$$

In the formula: *C* was the extinction coefficient, with a value of 0.60; *G* was the soil heat flux (W·m^{-2}); R_n was the net radiation (W·m^{-2}), and R_{ns} was the net radiation absorbed by the soil (W·m^{-2}), which was calculated based on the LAI extinction principle [19].

2.3.2 Structural Equation Modeling (SEM)

Structural Equation Modeling (SEM) was a method based on the covariance matrix that simultaneously analyzed the causal relationships between multiple variables. It integrated factor analysis and path analysis, allowing for the establishment, estimation, and testing of causal relationships in models. SEM could analyze the total, direct, and indirect effects of independent variables on dependent variables, making it suitable for studying the inter relationships between multiple variables. Typically, SEM used maximum likelihood estimation for parameter estimation, and the fit function of maximum likelihood estimation conformed to the χ^2 distribution. Therefore, in SEM, the χ^2 test was often used to evaluate model performance. The software Amos 24 was used

to run SEM, and the model was adjusted and revised based on the test results to analyze the control mechanisms of plant transpiration (*T*) and soil evaporation (*E*).

2.4 Calculation of model improvement parameters

The S-W model included a large number of parameters, in addition to environmental parameters, the core of its formula was the calculation and validation of five resistance parameters: r_s^s, r_a^s, r_s^c, r_a^c, and r_a^a. Based on the research findings of other scholars and combined with the actual conditions of the study area, the research methods for the three parameters raa, rac, and ras were relatively unified. This study followed the original model method and still used the formulas from reference [10] for calculation. Additionally, the two parameters r_s^c and r_s^s adopted the methods recommended by other scholars, and the calculation methods were as follows.

2.4.1 Vegetation canopy stomatal resistance(r_s^c)

$R_s^c$was expressed as a function of solar radiation, water vapor deficit, air temperature, and soil moisture content, according to the Jarvis method [21], namely:

$$r_s^c = \frac{r_{smin}}{LAI}\frac{F_1}{F_2F_3F_4} \tag{7}$$

$$F_1 = \frac{1+f}{f+\frac{r_{smin}}{r_{smax}}} \tag{8}$$

$$f = \frac{0.55R_{si}}{R_{sl}}\frac{2}{LAI} \tag{9}$$

$$F_2 = 1 - 0.0025D \tag{10}$$

$$F_3 = 1 - 0.0016(298 - T) \tag{11}$$

$$F_4 = \begin{cases} 1 & \theta r > \theta wp \\ \dfrac{\theta r - \theta wp}{\theta fc - \theta wp} & \theta fc \le \theta r \le \theta wp \\ 0 & \theta r < \theta wp \end{cases} \tag{12}$$

In the formula: r_{smin} represented the minimum stomatal resistance of the plant canopy (s·m^{-1}), and r_{smax} represented the maximum stomatal resistance of the plant canopy (s·m^{-1}). According to the research results of Cui Jianguo [22], they were taken as 170 s·cm^{-1} and 1400 s·cm^{-1}, respectively; F_i (1, 2, 3, 4) denoted the stress functions for specific environmental variables; R_{sl} was the shortwave radiation absorbed by the plant (W·m^{-2}); R_{si} was the solar radiation at the onset of photosynthesis

($W \cdot m^{-2}$); T_a was the air temperature (K); θ_r represented the soil moisture content at the plant root zone ($cm^3 \cdot cm^{-3}$, in this paper, the soil moisture content at a depth of 100~120 cm was used as a substitute); θ_{fc} was the field capacity ($cm^3 \cdot cm^{-3}$, usually taken as $0.75\theta_{sat}$); θ_{wp} was the plant wilting coefficient (0.05 $cm^3 \cdot cm^{-3}$);

2.4.2 Soil surface resistance(r_s^s)

r_s^s adjusted the movement of water vapor from the interior of the soil to the soil surface and had a strong dependence on the moisture content of the upper soil layer. Studies have shown that there was a functional relationship between r_s^s and the soil surface moisture content θ [23]. That is,

$$r_s^s = a_1\left(\frac{\theta_{sat}}{\theta}\right)^{b_1} + c_1 \quad (13)$$

In the formula: θ represented the soil surface moisture content ($cm^3 \cdot cm^{-3}$). Due to the complexity of the soil surface, the soil moisture content at a depth of 10 cm was used as a substitute in this paper. θ_{sat} was the soil saturated moisture content (0.41 $cm^3 \cdot cm^{-3}$), and a_1, b_1, c_1 were parameters that needed to be calibrated in the model.

2.5 Model evaluation methodology

The purpose of model evaluation was to determine whether the model met the research requirements. In this paper, the Root Mean Square Error (RMSE), Coefficient of Determination (R^2), and Nash-Sutcliffe Efficiency Coefficient (NS) were used to evaluate the accuracy of the model simulation. The calculation formulas and evaluation criteria are detailed in reference [2].